The Impact of Target Width and Amplitude on Task Performance and Error Rate: A Two-Way ANOVA Analysis

Team Information:

Rifat Rahman Khan[rkhan11@stevens.edu]

Srikala Sandu[ssandu1@stevens.edu]

Rachel Daniel[rdaniel@stevens.edu]

Study Design:

This study employs a within-subjects design to analyze task performance, with each participant completing multiple conditions under varying target sizes and distances. The independent variables are target size and target distance, while the dependent variables are mean completion time and error percentage. The study setup included:

Conditions: Four specific configurations combining target amplitude and width:

C0: Width = 25 , Amplitude = 200

C1: Width = 25 , Amplitude = 400

C2: Width = 50 , Amplitude = 200

C3: Width = 50 , Amplitude = 400

Trials and Blocks: Each condition was tested in five sessions (S1, S2, S3, S4, S5) per participant, totaling four participants.

Each participant performed tasks in a controlled setting designed to minimize external variables and maintain focus on the task parameters.

Four voluntary participants were recruited, each providing informed consent. These participants are graduate students from Stevens Institute of Technology, all specializing in fields related to computer science and pursuing master's degrees. The participants, ranging in age from 22 to 25 and are all male, South Indian, and fluent in English, with bachelor's degrees in computer science. Their technical background and significant computer experience ensured consistency in the study's baseline conditions. On average, each participant took 30 minutes to complete the study tasks.

Study Results:

In this study we analyzed the effects of target width and amplitude on task performance focusing on Mean Completion Time and Mean Click Error across four experimental conditions. The conditions were defined by combinations of width (low and high) and amplitude (low and high). The Figure-1 and Figure-2 demonstrates the Mean Completion Time and Mean Click Error for the four different condition codes. We can see that the condition code C2 (high width and low amplitude) has the lowest Mean Completion Time whereas, the condition code C2 (low width and high amplitude) has the highest. From Figure-2 we can see that the condition code C1 has the most Mean Click Errors. On the other hand, C3 has the least Mean Click Errors.

To statistically analyze the impact of width and amplitude on Mean Completion Time and Mean Click Errors, we conducted a Two-Way ANOVA. These statistical tests were performed using Python.

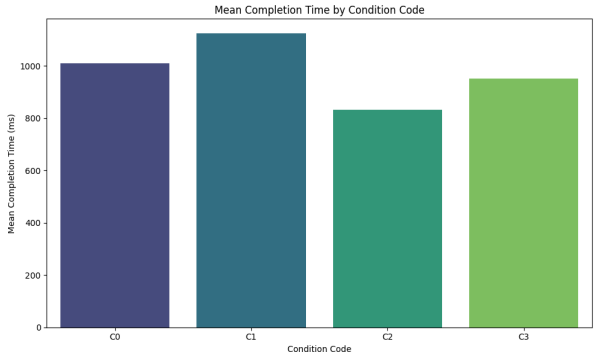


Figure-1

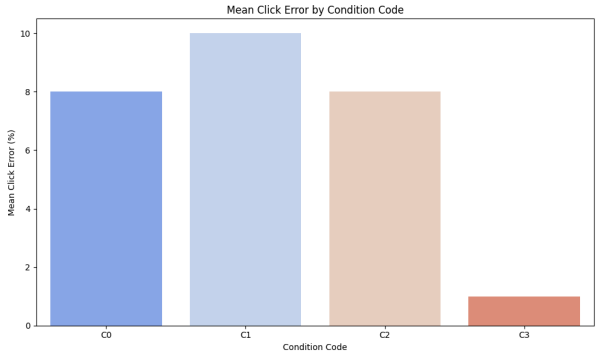


Figure-2

	A	B	C	D	E	F	G	H	I
1	Participant Code	Session Code	Condition Code	Hand Dominance	Pointing Device	Device Experience	Mean Completion Time (ms)	Mean Click Error (%)	Mean Throughput (bps)
2	P1	S1	C0	Dominant	Mouse	Intermediate	777.26	0	4.456472414
3	P1	S1	C1	Dominant	Mouse	Intermediate	901.9	20	4.374086475
4	P1	S1	C2	Dominant	Mouse	Intermediate	621.72	0	4.90724489
5	P1	S1	C3	Dominant	Mouse	Intermediate	829.54	0	3.793518375
6	P1	S2	C0	Dominant	Mouse	Intermediate	806.78	0	5.321731264
7	P1	S2	C1	Dominant	Mouse	Intermediate	913.98	20	3.783653729
8	P1	S2	C2	Dominant	Mouse	Intermediate	598.36	20	3.608476069
9	P1	S2	C3	Dominant	Mouse	Intermediate	738.34	0	4.007004604
10	P1	S3	C0	Dominant	Mouse	Intermediate	687.32	0	7.175288844
11	P1	S3	C1	Dominant	Mouse	Intermediate	958.84	20	4.058475773
12	P1	S3	C2	Dominant	Mouse	Intermediate	553.2	0	5.530889826
13	P1	S3	C3	Dominant	Mouse	Intermediate	722.38	0	4.117908829
14	P1	S4	C0	Dominant	Mouse	Intermediate	709.5	0	4.130436774
15	P1	S4	C1	Dominant	Mouse	Intermediate	828.04	20	4.964314121
16	P1	S4	C2	Dominant	Mouse	Intermediate	610.26	0	3.303182372
17	P1	S4	C3	Dominant	Mouse	Intermediate	852.3	0	4.7784315
18	P1	S5	C0	Dominant	Mouse	Intermediate	675.62	0	5.637325243

Dataset

For Mean Completion Time, the hypotheses were as below:

ANOVA Table for Mean Completion Time

Source	Sum of Squares	Degrees of Freedom	F-Statistic	P-Value
C(Width)	622085.2	1	10.13	0.002114
C(Amplitude)	271063.0	1	4.41	0.038961
C(Width):C(Amplitude)	204.4	1	0.0033	0.954143
Residual	4667170.0	76	nan	nan

Figure m.1

- Null Hypothesis (H0): There is no significant impact of width, amplitude or their interaction on Mean Completion Time
- Alternative Hypothesis (H1): There is a significant impact of width, amplitude or their interaction on Mean Completion Time

The results we found in our experiment suggested that width (p=0.002) and amplitude (p = 0.039) had significant impacts on Mean Completion time. These results reject the Null Hypothesis for these factors.

In this case, the interaction between width and amplitude was not statistically significant (p = 0.954). So, the combination of width and amplitude does not have an additional effect on task completion time beyond their individual effects.

For the Mean Click Error, the hypotheses were as follows:

ANOVA Table for Mean Click Error

Source	Sum of Squares	Degrees of Freedom	F-Statistic	P-Value
C(Width)	405.0	1	2.29	0.134057
C(Amplitude)	125.0	1	0.71	0.402782
C(Width):C(Amplitude)	405.0	1	2.29	0.134057
Residual	13420.0	76	nan	nan

Figure m.1

- Null Hypothesis (H0): There is no significant impact of width, amplitude or their interaction on Mean Click Error

- Alternative Hypothesis (H1): There is a significant impact of width, amplitude or their interaction on the Mean Click Error

Based on these findings we can say that the effect on Mean Completion Time of width and amplitude that we found in our experiment aligns well with Fitts Law. When the width of the target is high and the amplitude is low, we get the lowest Mean Completion Time. However, they do not have an additional effect on the Mean Click Error. If we had the opportunity to have more participants, we could have more reliable findings.

Reflection:

We learned how to collect data from participants and conduct statistical analyses, including hypothesis testing. Data cleaning was challenging, as participants sometimes entered incorrect condition codes. Patience was essential throughout the data collection process, as identifying and correcting minor errors proved difficult. Interestingly, different participants often yielded similar results. Overall, it was a valuable learning experience, and we also developed teamwork skills along the way.